SoungEil Houng

Ph.D. Candidate, Civil and Environmental Engineering, University of California, Berkeley

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Education in Civil Engineering

- **University of California, Berkeley** — Berkeley, CA
  Ph.D. in Civil and Environmental Engineering (Expected May 2027) — *Jan. 2024 – Present*
- **University of California, Berkeley** — Berkeley, CA
  M.S. in Civil and Environmental Engineering — *Jan. 2024 – May 2025*
- **New York University** — Brooklyn, NY
  (Transferred to UC berkeley) Ph.D. in Civil Engineering — *Aug. 2023 – Dec. 2023*

Education in Earth Sciences

- **Yonsei University** — Seoul, South Korea
  (All But Dissertation) Ph.D. in Earth System Sciences (Seismology/Geophysics) — *Sep. 2013 – Sep. 2015*
- **Yonsei University** — Seoul, South Korea
  M.S. in Geological Sciences (Seismology/Geophysics) — *Sep. 2011 – Aug. 2013*
- **Yonsei University** — Seoul, South Korea
  B.S. in Earth System Sciences — *Mar. 2006 – Aug. 2011*

Employment

- **Korea Hydro & Nuclear Power – Central Research Institute** — Daejeon, South Korea
  Seismic Hazard Analyst for Nuclear Power Plant Sites — *May 2017 – Aug. 2023*
- **National Institute of Meteorological Sciences** — Seogwipo, South Korea
  Seismologist for Seismic Discrimination of Explosion (Nuclear Experiment) — *Oct. 2015 – May 2017*

Peer-Reviewed Journal Publications

[1] **Houng, S. E.**, and Ceferino, L. (2026). Scalable Scenario-based Regional Earthquake Risk Assessment via Gaussianized Ground-motion–damage Modeling and PPCA, Accepted in *Earthquake Engineering & Structural Dynamics*. [link]

[2] **Houng, S. E.**, Ceferino, L., and Abrahamson, N. (2026). Fast Propagation of Epistemic Uncertainty in Seismic Hazard via Adaptive Importance Sampling, *Bulletin of the Seismological Society of America* **116**, 1709–1726. [link]

[3] **Houng, S. E.**, and Ceferino, L. (2025). Fast Probabilistic Seismic Hazard Analysis through Adaptive Importance Sampling, *Bulletin of the Seismological Society of America* **115**, 646–663. [link]

[4] **Houng, S. E.** (2018). Discrimination Between Natural Earthquakes and Explosions Based on the Azimuthal Distribution of S/P Amplitude Ratios, *Bulletin of the Seismological Society of America* **108**, 218–229. [link]

[5] **Houng, S. E.**, Lee, J., and Hong, T.-K. (2016). Dynamic Seismic Response of a Stable Intraplate Region to a Megathrust Earthquake, *Tectonophysics* **689**, 67–78. [link]

[6] Hong, T.-K., Park, S., and **Houng, S. E.** (2016). Seismotectonic Properties and Zonation of the Far-Eastern Eurasian Plate Around the Korean Peninsula, *Pure and Applied Geophysics* **173**, 1175–1195. [link]

[7] Hong, T.-K., Lee, J., and **Houng, S. E.** (2015). Long-Term Evolution of Intraplate Seismicity in Stress Shadows After a Megathrust, *Physics of the Earth and Planetary Interiors* **245**, 59–70. [link]

[8] Hong, T.-K., **Houng, S. E.**, and Jo, E. (2014). Temporal Changes of Medium Properties During Explosive Volcanic Eruption, *Geophysical Research Letters* **41**, 1944–1950. [link]

[9] **Houng, S. E.**, and Hong, T.-K. (2013). Probabilistic Analysis of the Korean Historical Earthquake Records, *Bulletin of the Seismological Society of America* **103**, 2782–2796. [link]

Patents

[1] **Houng, S. E.**, and Ceferino, L. (University of California, Berkeley) (2024). Computational Framework for Numerical Probabilistic Seismic Hazard Analysis (PSHA). U.S. Provisional Application No. 63/664096; International Application No. PCT/US2025/034042 (*Pending*). [link]

[2] **Houng, S. E.**, Kim, H.-S., and Park, D. (Korea Hydro & Nuclear Power Co., Ltd.) (2020). Method for Automatically Discriminating Earthquake Vibratory Motion from Erroneous Signal Record at Seismic System of Nuclear Power Plant. Registered Oct. 5, 2023 (Registration No. 1025879120000). [link]

Invited Talks

- "Scalable Scenario-based Earthquake Risk Modeling via Linearized Ground-Motion–Fragility Coupling and Probabilistic PCA," *NHERI GSC RSR Meeting* Feb. 2026
- "Adaptive Importance Sampling based PSHA Computational Framework," *Korea Hydro & Nuclear Power – Central Research Institute*, Daejeon, South Korea Jul. 2025
- "PSHA: Theory, Practice, and Numerical Computation," *Korea Institute of Geosciences and Mineral Resources*, Daejeon, South Korea Apr. 2025
- "Current State-of-the-Art PSHA for Korea's NPPs," *Korea Atomic Energy Research Institute*, Daejeon, South Korea Jul. 2020
- "Calculation of Probabilistic Seismic Hazard Analysis," *Korea Institute of Civil Engineering and Building Technology*, Goyang-si, South Korea May 2019
- "Evaluation of the Ground Motion at NPP Sites using Empirical Green's Function," *Korea Institute of Nuclear Safety*, Daejeon, South Korea Apr. 2019
- "Development of PSHA Calculation Program," *2018 Site-Specific GMRS Project Workshop*, POSTECH, Pohang, South Korea Dec. 2018
- "Synthesis of Strong Ground Motion using Empirical Green's Function," *2018 KHNP-CRIEPI-KEPRI General Meeting*, Seoul, South Korea Nov. 2018
- "Synthesis of Strong Ground Motion using Empirical Green's Function," *2018 Fault Source Modeling Workshop*, KEPRI, Daejeon, South Korea Aug. 2018
- "Synthesis of Strong Ground Motion using Empirical Green's Function," *2018 Nuclear Safety & Security Information Conference*, Daejeon, South Korea Mar. 2018
- "Synthesis of Strong Ground Motion using Empirical Green's Function," *Korea Institute of Geosciences and Mineral Resources*, Daejeon, South Korea Feb. 2018

Experience

- **GSR**, *UC Berkeley* Spring 2024 – Present
- **GSI**, CE 193 Engineering Risk Analysis, *UC Berkeley* Fall 2026
- **GSR**, AI4CEE: Integrating AI in CEE Undergraduate Educaion, *UC Berkeley* Spring 2026
- **GSI**, CE 294 Disaster Risk Analysis of Infrastructure Systems, *UC Berkeley* Fall 2024
- **TA**, Computing Geology and Lab, *Yonsei University* Fall 2014
- **TA**, Origin and Evolution of Earth, *Yonsei University* Spring 2014
- **TA**, Geophysical Exploration and Lab, *Yonsei University* Fall 2011, Fall 2013
- **TA**, Earth and the Universe, *Yonsei University* Spring 2013
- **TA**, Geophysics, *Yonsei University* Spring 2012

Technical Skills

- **Programming Languages**: C/C++, Python, Fortran, MATLAB, Bash shell, Awk
- **Seismic Hazard/Risk Tools**: Openquake, R2D, OpenSHA, HAZ45, FRISK88
- **Seismological Tools**: SMSIM, SAC, TauP, Coulomb
- **Plotting Tools**: Matplotlib, Seaborn, Gnuplot, Generic Mapping Tools (GMT)
- **Operating Systems**: Linux (Red Hat, CentOS), MS Windows